GEK-4798A

INSTRUCTIONS

3S7931NA143B

VOLTAGE REGULATOR EQUIPMENT FOR ALTERREX* SYSTEM

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GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

3S7931NA143B

VOLTAGE REGULATOR EQUIPMENT

INTRODUCTION

The 3S7931NA143B Voltage Regulator equipment is designed to regulate automatically the terminal voltage of alternating current generators and synchronous condensers. An amplidyne generator, which is used as the output stage of the regulator, controls the voltage of the exciter, which in turn controls the voltage of the AC machine. A functional block diagram is shown in Figure 19.

The regulator is normally equipped with an under-excited reactive ampere limit circuit. This circuit prevents the excitation of the AC machine from being automatically reduced materially below the desired minimum value which is either dependent upon or independent of the load on the AC machine. The regulator and the under-excited reactive ampere limit components are located on separate panels.

A reactive current compensator is included to permit proper division of reactive current, and to restrict the flow of cross currents between paralleled machines. In the case of single machines, its connections may be reversed to reduce the effective impedance of transformer banks. An active and reactive current compensator is occasionally provided to facilitate the regulation of voltage at some point remote from the machine terminals. Combinations of the two compensators can be furnished which will accomplish both functions simultaneously. A motor starter for the amplidyne drive motor, a magnetic switch for placing the amplidyne in the exciter field circuit, a meter panel, a switchboard type amplidyne voltmeter and the necessary control transformers, switches and relays are included with the regulating equipment.

Both the number and the arrangement of auxiliary components vary with the requirements of the particular application. Therefore, it is necessary to refer to the material lists and diagrams provided for each individual application to determine what material is included and what connections should be used. The diagrams shown in this book are for illustrative purposes only; they are not intended to apply to all installations.

RECEIVING, HANDLING AND STORAGE

RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked, it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress.

DESCRIPTION

COMPONENT ARRANGEMENTS

The 3S7931NA143B central station regulator is generally housed in a floor-mounted enclosure that can be opened at both the front and the back. It usually includes the following units:

1. Voltage regulator panel
2. Regulator amplifier panel
3. Under-excited reactive ampere limit panel
4. Meter panel
5. DC control panel which may contain a transfer contactor, current limiting resistors, fuses and other control devices.
6. AC control panel which may contain a motor starter, a fused safety switch, an undervoltage relay and other control devices.
7. An amplidyne M-G set with flywheel and permanent magnet alternator overhung on the amplidyne.

All of the above panels have steel bases with the possible exception of the DC control panel which may have a base of a compound material.

The amplidyne is mounted either in a separate location from the regulator enclosure or in an adjoining enclosure which also includes the exciter field rheostat.

VOLTAGE REGULATOR

General

As shown in Figure 20 the regulator can be placed in,

or removed from, service by operating the regulator control switch.

With the control switch at OFF, the excitation and terminal voltage of the AC machine can be manually controlled by adjusting the exciter-field rheostat. Under this condition, the regulator is completely out of service and the amplidyne generator armature is disconnected from the exciter-field circuit.

With the control switch at ON, the regulator will control automatically the excitation and terminal voltage of the AC machine.

When the AC machine voltage is higher than normal, the regulator will supply control-field current to the amplidyne, so that the amplidyne voltage will oppose, or "buck", the exciter voltage impressed across the exciter-field circuit. This will tend to lower the exciter voltage and decrease the AC machine field current and terminal voltage.

When the AC machine terminal voltage is below normal the regulator will supply control-field current to the amplidyne which will produce an amplidyne voltage in the opposite direction to raise, or "boost", the exciter voltage impressed across the exciter-field circuit. This will tend to raise the exciter voltage, increase the AC machine field current, and raise the terminal voltage of the AC machine toward normal.

The manual control can be manually adjusted so that at any given load the resistance of the exciter-field circuit will be correct for normal AC machine voltage, and the amplidyne voltage will be zero. If the load on the AC machine is increased, the resultant drop in terminal voltage will cause the regulator to produce an amplidyne voltage in the boost direction to raise the AC machine excitation. As the load on the AC machine is still further increased, the amplidyne voltage will increase further in the boost direction to continually raise the excitation in an effort to maintain normal terminal voltage. Should the load be decreased to the original load value, the machine excitation requirements will be less, and the amplidyne boost voltage will fall until it becomes zero. Lowering the load on the AC machine beyond this value will cause the regulator to produce an amplidyne voltage in the buck direction to reduce the AC machine excitation.

The regulator control switch is provided with a position to be used for test and adjustment of the regulator preparatory to placing it in service. With the control switch at TEST, the regulator is connected to the signal-voltage source and the transfer contactor is so positioned that the amplidyne will operate independently without affecting the excitation of the AC machine. This permits operation of all regulator equipment for test purposes.

The regulator control switch may also be provided with INTER (intermediate) and POLARIZE positions for correcting the polarity of the exciter armature if the polarity is the reverse of that required by the voltage regulator. With the control switch at INTER, the AC motor-amplidyne generator set is started. With the control switch at POLARIZE, the polarity of the exciter is corrected. In this latter position the regulator is disconnected from the signal-voltage source, the transfer contactor is so positioned that the amplidyne armature is connected in the exciter field circuit, and the amplidyne is energized to provide proper voltage and polarity to the exciter field.

CAUTION

The polarity of the exciter should be corrected only when the AC machine is not synchronized with the power system or when the exciter is not connected to the AC machine field.

Regulator Signal Voltage

When the control switch is in the TEST and ON positions, the regulator is connected to the secondaries of potential transformers whose primaries are connected to the AC machine terminals or bus. These transformers supply a three-phase voltage to the regulator which is directly proportional to the machine terminal voltage. The regulator input signal consists of this voltage plus the vectorial addition to one phase of voltage from the reactive-current compensator as shown in Figure 1. The compensator is described in greater detail in a following section.

Regulator Comparison and Amplifier Circuit

The regulator comparison and amplifier circuit illustrated in Figure 1 takes an input voltage from potential transformers from the generator, compares this to a reference voltage, and furnishes an output error signal to drive push-pull buck-boost magnetic amplifiers.

The input voltage from the PT's is reduced from 115 volts to approximately 24 volts by transformer A1T and A2T and rectified by diodes 1D through 6D to provide a DC voltage across circuits 19 to 29 that is proportional to generator terminal voltage. A portion of this voltage, across circuits 2 to 29, is applied to 1Q transistor. Transistors 1Q and 2Q act as a summing amplifier. The input to transistor 2Q is the reference voltage developed across 1Z Zener diode. The output of 2Q transistor is the difference between the feedback signal applied to transistor 1Q and the reference voltage applied to transistor 2Q. This difference or error signal appears across 5R resistor. This error signal is reversed in sense by 3Q transistor, then amplified by 4Q and 5Q transistors to provide an amplified error signal across 12R resistor.

This amplified error signal is then applied to 6Q and 7Q transistors to provide a voltage to drive the regulator magnetic amplifiers. This final output voltage appears across 22R.

Should the current limit circuit override the regulator circuit, then transistor 6Q will be controlled by the current limit circuit (26). The output voltage

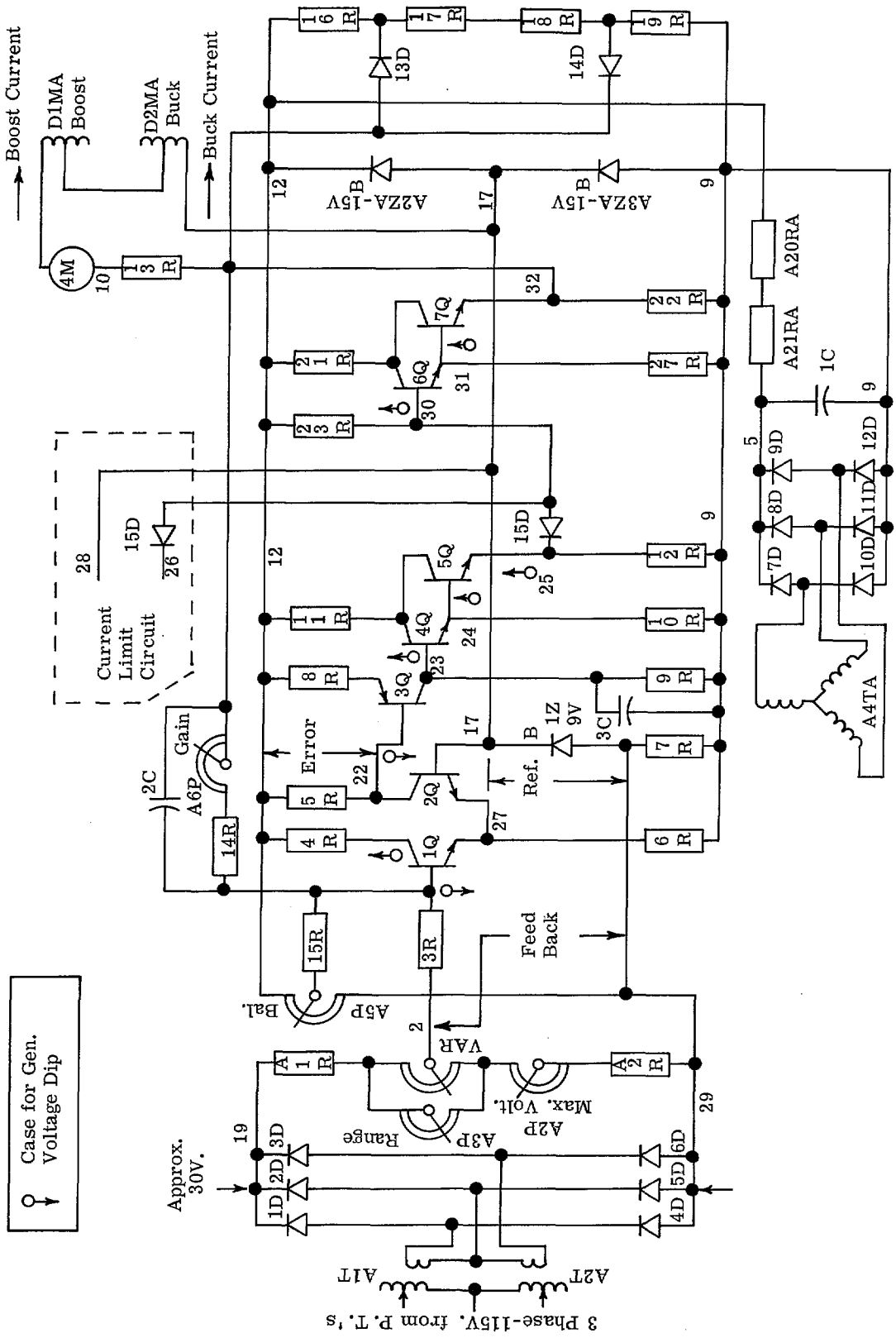


Figure 1. Regulator Comparison and Amplifier Circuit

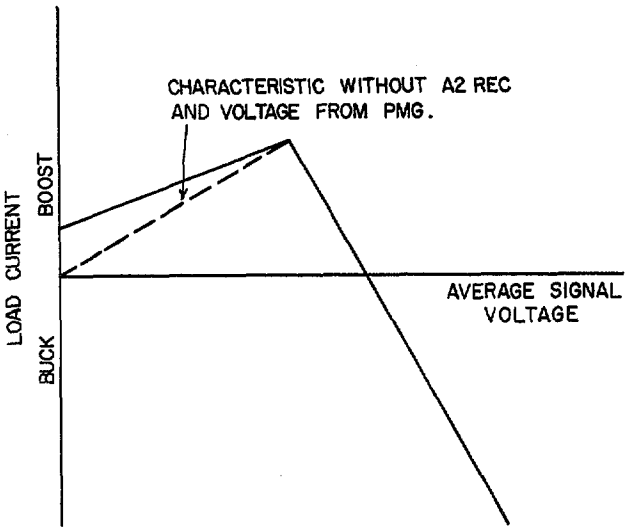


Figure 2. Load Current-Signal Voltage Characteristic of Voltage Regulator

(32) is fed back through the gain adjustment to compensate for tolerances in the various transistors. Capacitors 2C and 3C act as filters to reduce ripple.

Transformer A4TA furnishes power for the entire transistor circuit. Zener diodes A2ZA and A3ZA furnish regulated busses of +15 volts and -15 volts with respect to the common circuit 17.

During normal operation should generator terminal voltage dip, due to increased load, then the base of transistor 1Q will go negative. The error signal (22) will go negative. The input to 4Q transistor (23) will go positive. Circuit 25 will go positive. Transistors 6Q and 7Q will follow circuit 25 causing circuit 32 to also go positive. This will cause more boost current to flow through magnetic amplifiers D1 and D2, thereby increasing excitation to the system to correct for the original voltage dip.

Reactive Current Compensator (RCC)

The RCC is used to equalize division of reactive kva and to prevent circulating reactive current between AC machines when two or more machines with individual regulators are operating in parallel. As shown in Figure 3, the adjustable reactor winding is connected to the secondary of a current transformer located in one of the AC machine lines, while the insulating winding is connected on one phase of the three-phase regulator signal voltage. IT IS ESSENTIAL THAT BOTH WINDINGS OF THE COMPENSATOR BE CONNECTED IN THE SAME PHASE.

The compensator will vary the regulator signal voltage in this phase mainly as a function of the machine reactive current. As the machine overexcited reactive current increases, the voltage will be increased. As a result, the average three-phase signal voltage supplied to the regulator will rise. Due to this higher voltage signal, the regulator will act to reduce excitation, thereby reducing the over-excited reactive current.

If the AC machine is operating in the underexcited region, the compensator will decrease the average three-phase voltage signal presented to the regulator, so that the regulator will act to increase the excitation and decrease the underexcited reactive current. The effect of the compensator can be increased by increasing the active reactance of the unit appearing in the current-transformer secondary circuit. This may be done readily with the tap switches located on the reactor.

If two AC machines are connected through transformers, the compensator is not normally used as described above, instead one winding is reversed to obtain the opposite effect to that described above. This makes it possible to adjust the transformer impedance to an optimum value of 6 or 7 percent on a reactive basis, improving system voltage regulation.

The compensator is adjustable in one-volt steps, on the basis of five amperes in the current-transformer secondary, by means of two manually operated tap switches. One of the separately adjustable tap switches gives coarse adjustments and the other fine. The voltage across this winding in the regulator signal voltage circuit is proportional to the current flowing through the current transformer secondary winding. With five amperes flowing through the compensator and with the reactance all cut in, the total drop is 24 volts. This is equivalent to about 36 percent line-to-neutral voltage drop in one signal lead to the regulator, or an average effect of approximately 12 percent line-to-neutral voltage drop on a three-phase basis at zero power factor.

The compensator is built with two independent windings, as shown in Figure 3, so that the current and potential transformer secondaries are electrically separate and both may be grounded.

Regulator Adjustments

The regulator has fixed adjustments in addition to the variable voltage adjusting rheostat. Slide wire resistor A4R may be used to lower regulator sensitivity by increasing its resistance. Tap connections to the open delta input transformers may be changed if the normal incoming signal voltage is not 115 volts. Resistors A1R and A2R determine the range of voltage adjustment.

Regulator Amplifier

The circuit of a single magnetic amplifier is shown in Figure 4, and its characteristic curve is shown in Figure 5. As shown in Figure 5, zero control current results in a load current of very nearly the maximum value. A bias current in one of the control windings shifts the characteristic curve to the right so that the load current may be increased or decreased

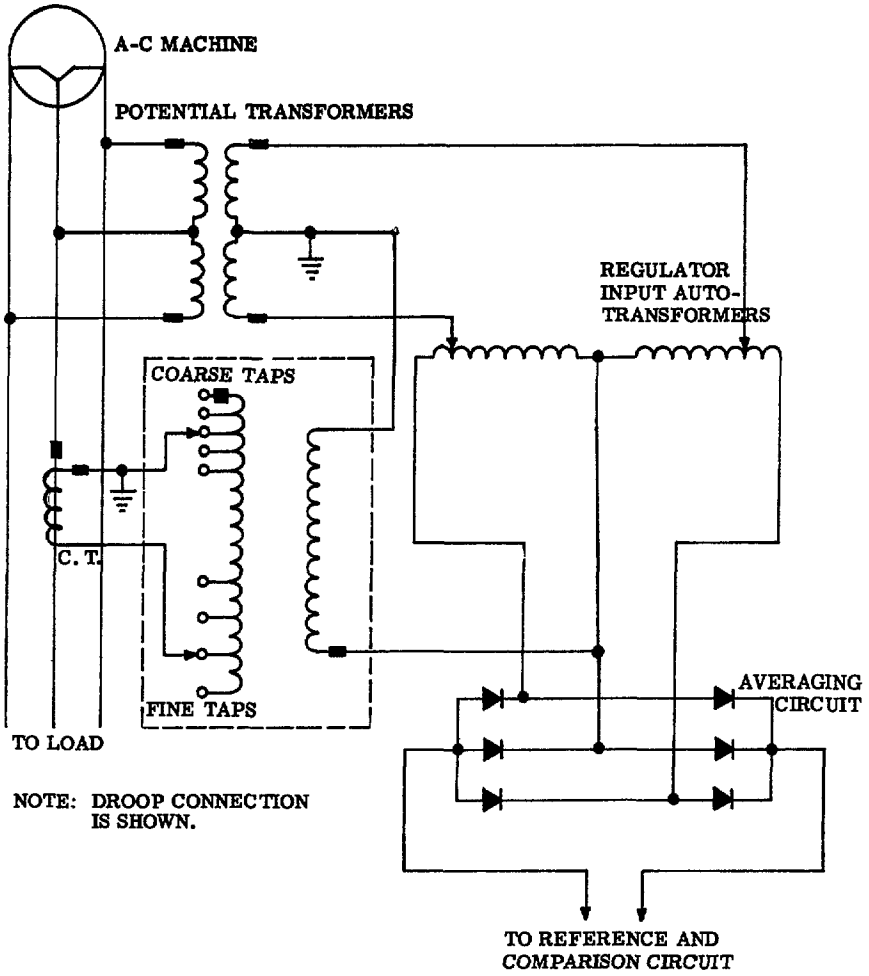


Figure 3. Reactive Current Compensator Connections

by approximately equal values for equal positive or negative signal inputs. The zero signal load current flows in a control field of an amplidyne generator and its effect on the amplidyne is cancelled by an equal but opposite current flowing in a second amplidyne control field. This equal and opposite current is obtained from a second magnetic amplifier whose main control winding is connected in parallel with the control winding of the first amplifier but with opposite polarity. This push-pull combination of magnetic amplifiers shown in Figure 20 has the characteristic shown in Figure 6. The net change in amplidyne control field current for a given input signal to the amplifiers is about twice that which is obtained from a single amplifier. Scott T transformers supply 2-phase, 200 cycle power to the amplifiers from the 3-phase PMG. In addition to the main amplifier input from the regulator panel and the bias input mentioned above, there are also inputs from the underexcited reactive ampere limit amplifier, exciter voltage, and sometimes from amplidyne armature voltage and current as shown in Figure 20. The underexcited reactive ampere limit and its associated amplifier are described in detail in a following section. The exciter and amplidyne voltage inputs are used to stabilize the regulator.

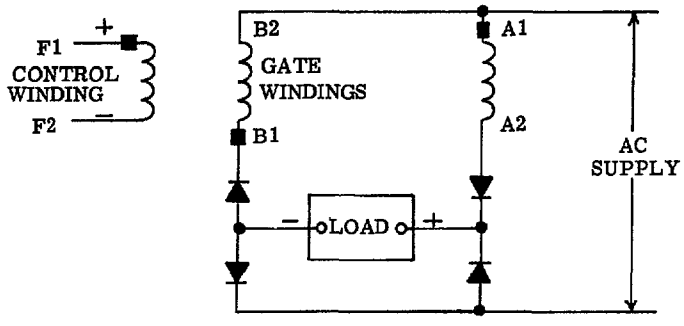


Figure 4. Elementary Diagram of a Magnetic Amplifier

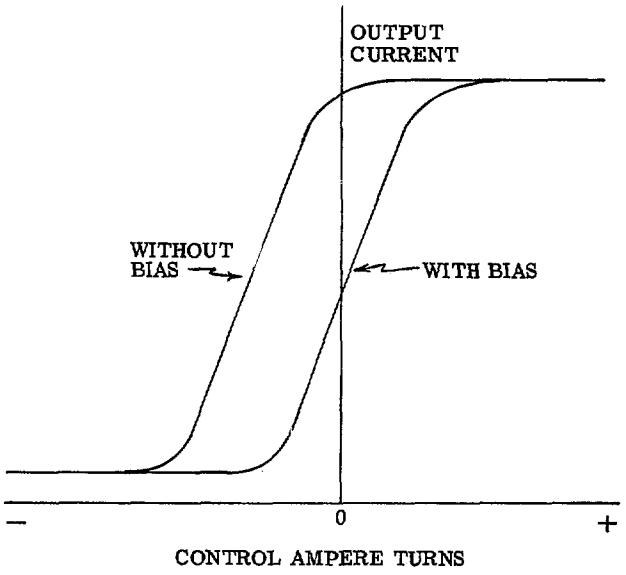


Figure 5. Magnetic Amplifier Characteristic Curve With and Without Bias

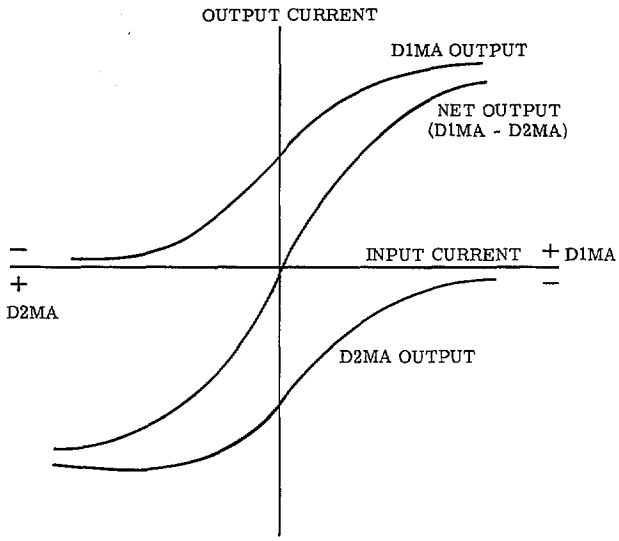


Figure 6. Characteristic Curves of Push-Pull Magnetic Amplifier

Only transient changes in these voltages can cause current to flow in the control windings since they are connected to the amplifier through series capacitors. These transient currents act in a direction to oppose the voltage changes which caused them, damping the transients and preventing sustained oscillations.

Regulator Amplifier Adjustments

The push-pull amplifiers have bias and load resistance adjustments which are set in the factory for optimum performance of the amplifiers, and stabilization adjustments, which are also set in the factory, for optimum over-all regulator performance. It may be necessary to make slight changes in the stabilization adjustments at the time of installation as described in the "Installation and Adjustment" section of this book.

Regulator Operation

When the terminal voltage of the AC machine is equal to the value which the regulator is adjusted to maintain, no current will flow through the control windings of the amplifiers. This results in equal currents through the two control fields of the amplidyne, and zero output voltage from the amplidyne.

An increase in AC machine voltage above the value for which the regulator is adjusted to maintain, will result in a bucking current from the comparison circuit flowing through the control windings of the amplifiers from D2 to D1. As a result the current output of D2MA increases and the current output of D1MA decreases. The current through amplidyne field F4-F3 is then greater than the current through F1-F2. Amplidyne terminal A1 then becomes positive with respect to C2. This causes a bucking current in the exciter field, a reduction in exciter voltage, and a decrease in AC machine voltage.

Similarly, a decrease in AC machine terminal voltage below that for which the regulator is adjusted will result in a boosting current from the comparison circuit into the amplifier from D1 to D2, and a boosting voltage from the amplidyne. This will produce an increase in AC machine excitation and an increase in terminal voltage.

The currents which flow as a result of stabilizing circuits are in the opposite direction to those described above.

The exciter voltage stabilizing circuit consists of resistor D5R, rheostat D3P, capacitor D3C, and series connected control windings on the magnetic amplifiers. An increase in exciter voltage causes a current to flow through these components and the magnetic amplifier from terminal D6 to D5 to charge capacitor D3C to the higher exciter voltage. This current increases the output from buck amplifier D2MA and decreases the output from boost amplifier D1MA. The amplidyne voltage changes in the buck direction decreasing exciter voltage. This decrease, in opposition to the original increase, slows the rate of change of exciter voltage and stabilizes the regulator. The amplidyne stabilizing circuit, when furnished, consists of resistor D6R, rheostat D4P, capacitor D4C, and another series connected pair of control windings on the magnetic amplifiers. This circuit works in a similar manner to the exciter voltage stabilizing circuit, except that it responds to changes in amplidyne voltage.

Since in both cases the magnitude of the signals fed back into the amplifiers is dependent upon the rate of change of the stabilizing voltage, this method of stabilization is known as rate feedback.

The amplidyne may have a negative feedback circuit consisting of resistor D4R and amplidyne control field F5-F6 connected across the amplidyne armature terminals A1-C2. A positive increase in voltage C2 with respect to A1 causes a current to flow from F6 to F5 decreasing the amplidyne voltage. The over-all effect is to decrease amplidyne gain, reduce sensitivity and increase stability of the regulator.

Underexcited Reactive-Ampere Limit (U. R. A. L.)

The function of the underexcited reactive-ampere limit circuit, shown in Figure 20, is to prevent the regulator from reducing the AC machine excitation to such a low value that the machine would fall out of synchronism. It operates by limiting the machine underexcited reactive current.

Under normal operating conditions, with the regulator and the limit in service, the underexcited reactive current of the AC machine is less than the value at which the limit functions, and the limit is inoperative. However, if the regulator reduces the machine excitation to a sufficiently low value, the limit becomes operative and prevents the underexcited reactive current from rising materially above the value for which the limit is adjusted. The direct-current output from the limit circuit is amplified and applied to the limit amplifier which recalibrates the voltage regulator to hold a higher machine voltage. Consequently, the limit will prevent appreciable increases in machine underexcited reactive current beyond a predetermined value.

Since, in many cases, the magnitude of the underexcited reactive current at which the AC machine would fall out of synchronism is a function of machine active current, the limit can be adjusted so that the value of underexcited reactive current at which the limit will become operative (the limit-start point) will decrease as the machine active current increases. Therefore, it is possible to arrange the limit so that its starting point is automatically changed as machine active current varies. Active current is the in-phase or watt component of the machine armature current.

Figure 7 illustrates the loss of stability characteristic of a typical AC machine, as well as a typical setting of the underexcited reactive ampere limit. The solid curve represents the limit of stable operation of the AC machine while the dotted line indicates a particular setting of the underexcited reactive ampere limit. The loss of stability characteristic is a function of the AC machine and the power system and must be determined for each case.

The solid curve demonstrates that with no load on the AC machine the underexcited reactive current which is allowable, without exceeding the steady-state stability limit, is greater than the allowable underexcited current at full load. To prevent loss of steady-state stability, the underexcited reactive ampere limit acts to prevent the generator from operating below the solid curve, and therefore must be adjusted to become effective at lower values of underexcited reactive current than those shown on the solid curve. This results in a setting as indicated by the dotted characteristic curve in Figure 7. The distance between the curves represents a margin between the underexcited reactive ampere limit operating point and the steady-state stability characteristic.

The loss of steady-state stability characteristic illustrated in Figure 7, and the underexcited reactive ampere limit start point, are both similar functions of terminal voltage of the AC machine. Therefore, a setting of the underexcited reactive ampere limit establishes the limit start curve for a particular terminal voltage on the machine which will protect against loss of steady-state stability for normal variations in the terminal voltage of the AC machine.

The underexcited reactive ampere limit circuits include the following:

1. Reactive ampere sensitive circuit.
2. Comparison circuit.
3. Limit amplifier circuit.
4. Stabilizing circuits.

These are shown with the voltage-regulator circuits in Figure 20.

Reactive Ampere Sensitive Circuit

The reactive ampere sensitive circuit obtains its input from one phase of the output of the alternator potential transformers, and one alternator current transformer.

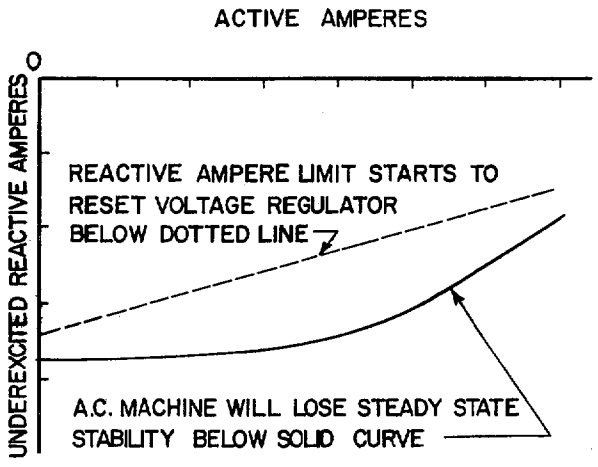


Figure 7. Typical AC Machine Pull-Out and Reactive-Ampere Limit-Start Characteristics

It is the function of this circuit to detect an increase of the reactive current in the alternator beyond a predetermined value in the underexcited direction, and to send a signal into the comparison circuit which ultimately acts to limit this increase of reactive current. Referring to Figure 20, the reactive ampere sensitive circuit consists of transformer B3T with a multitapped secondary, limit-start variable transformer B1VT, resistor B3R, reactive ampere limit power recalibration tap switch B1SW, reactor B3X, and isolating transformers B1T and B2T. The secondary of B3T has four terminals numbered 3, 4, 5 and 6. Terminal 4 is the midpoint between 3 and 5.

With limit-start variable transformer B1VT connected across terminals 4 and 6 of transformer B3T secondary, the limit start is adjustable in the underexcited reactive current region. With the variable transformer B1VT connected across terminals 3 and 4 of transformer B3T, the limit start is adjustable in the overexcited reactive current region. Since this latter connection is not normally used, it will not be discussed in detail. However, limit operation with this connection is similar to that for underexcited operation, and it may be used at the time of initial operation. When the POWER RECALIBRATION switch is at zero, the reactance in the circuit from reactor B3X is zero ohms, and the setting of LIMIT START variable transformer B1VT determines the reactive current at which the limit functions regardless of the power loading on the alternator.

The output voltages of the reactive ampere sensitive circuit are impressed on the primaries of isolating transformers B1T and B2T. It is an inherent property of the reactive ampere limit circuit that when the reactive current reaches the predetermined value where the limit action starts (the limit-start point) the voltages across B1T and B2T are equal in magnitude. An increase in reactive current in the underexcited direction above this predetermined value causes the voltage across B1T to exceed that across B2T.

A vector diagram of the reactive ampere sensitive circuit input quantity is shown in Figure 8. At unity power factor, the vector position of I_2 and, hence, the voltage drop across resistor B3R (in phase with and proportional to machine phase 2 current) is in quadrature with secondary voltage of B3T (in phase with and proportional to generator line 1 to line 3 voltage).

A vector diagram of reactive ampere sensitive circuit voltages for the case when the reactance from reactor B3X in the circuit is zero ohms, is shown in Figure 9. At unity power factor the voltage across B3R is represented by the vector Mb. For given overexcited and underexcited conditions, it is Ma and Mc, respectively. Under these conditions, the voltages on transformer B1T are 5B, 5A and 5c, respectively, and the voltages on transformer B2T are 3b, 3a and 3c, respectively.

In all these cases, the voltage on B2T exceeds that on B1T and the reactive ampere limit will be inoperative. When the current vector swings further in the underexcited direction so that the voltage across resistor B3R is Md, the voltage on B1T (5d) equals that on B2T (3d). Now any increase in underexcited reactive current will cause the voltage on B1T to exceed that on B2T which will cause the comparison circuit to excite the voltage resetting circuit and oppose further increases in underexcited reactive current. Therefore, the reactive ampere limit will operate to prevent vector Md from extending materially beyond the vertical limit line 4d. Thus, the underexcited reactive current tends to be limited proportionally to the vector length M4. By varying the position of M between 4 and 6 by moving the slider on variable transformer B1VT, it is possible to change the maximum value of underexcited reactive current at which the limit will operate. Since the active power component of current is in quadrature with the reactive current, the reactive current limited is independent of the amount of active power.

The purpose of the power recalibrating reactor (B3X) adjustment is to change the limit-start point as a function of the active current delivered by the machine. Figure 10 shows that when the machine is delivering active current, power recalibrating reactor B3X will cause the reactive ampere limit to become operative at a value of underexcited reactive current less than that set by the limit-start adjustment. With zero active amperes, the limit will start to reset the regulator with A underexcited amperes which is determined by the limit-start adjustment. As the active amperes increase, less underexcited reactive amperes

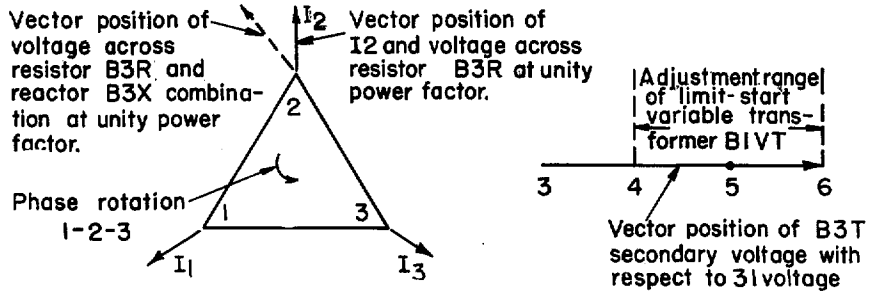


Figure 8. Vector Positions of Input Quantities of Reactive-Ampere Sensitive Circuit

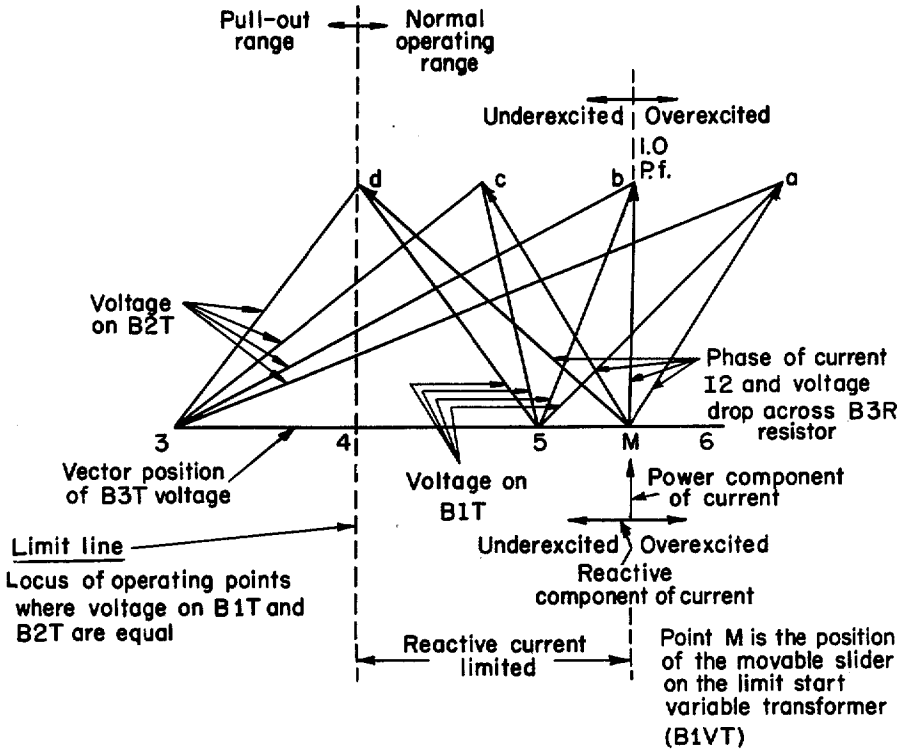


Figure 9. Vector Diagram for Reactive-Ampere Sensitive Circuit Without Power Recalibration Reactance in Circuit

are required to reset the voltage regulator.

If some of the reactance of B3X is connected in series with the resistance of B3R, the voltage drop across the resistance and reactance will be shifted in the underexcited direction as shown in Figure 8. The vector diagram will now be as shown in Figure 11. Three operating points are shown at b, c, and d, where the limit is ready to start. At point d there is no active power load on the machine. The value of the reactive current limit is proportional to the vector Md and is equal to that limited without the reactance of B3X in the circuit since the impedance in the resistor-reactor (B3R-B3X) circuit has been increased by the ratio of Md to M4. At point c, the power component of voltage across the resistor-reactor combination is Mc' and the reactive current limited in proportional to cc'. At point b, the power component of voltage across the resistor-reactor (B3R-B3X) combination is Mb' and the reactive current limited is proportional to bb'. Thus, the reactive current limited decreases as the active current increases. This amounts to a linear decrease of reactive amperes limited as a function of active amperes.

Comparison Circuit

The voltages from the secondaries of B1T and B2T are applied to the comparison circuit. The function of this circuit is to allow the flow of DC control current into the limit amplifier circuit, when the voltage across transformer B1T exceeds that across B2T.

The comparison circuit consists of bridge-type rectifiers B1REC and B2REC together with resistors B1R and B2R, capacitors B1C and B2C and reactors B1X and B2X, necessary for filtering and rectifier pre-

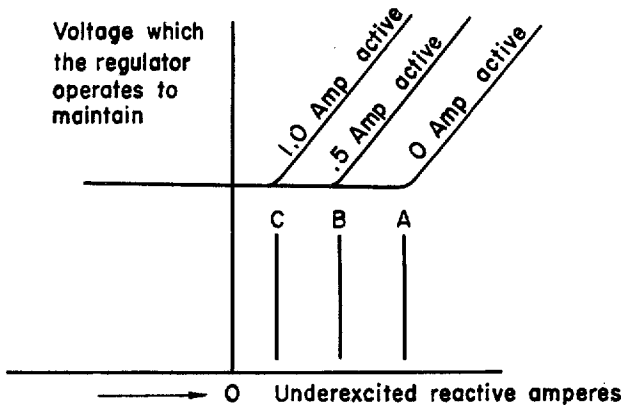


Figure 10. Reactive-Ampere Limit Power Recalibration Characteristic

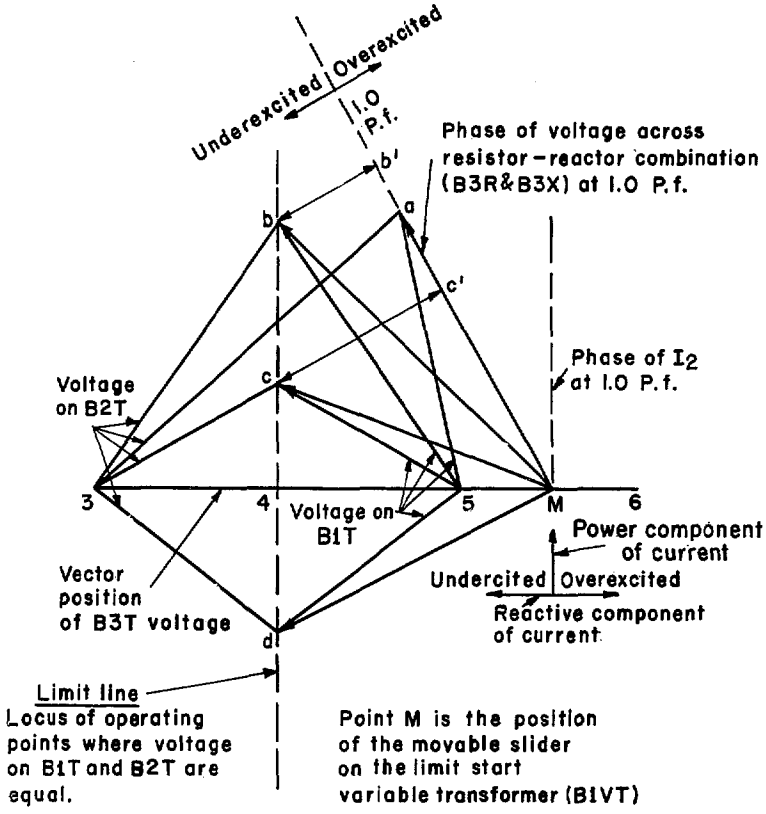


Figure 11. Vector Diagram for Reactive-Ampere Sensitive Circuit With Some Power Recalibration Reactance

loading. In addition, inverse voltage-limiting rectifier (B3REC) is used to allow a unidirectional flow of control current through the DC winding of the limit amplifier only when the voltage across B1T is greater than the voltage across B2T.

Limit Amplifier

The limit amplifier shown in Figure 20 obtains its input from the underexcited reacti e ampere limit described in the previous section. The limit amplifier output current flowing through series control windings of the regulator amplifier from D3 to D4 is a boost signal. This causes machine voltage to rise and underexcited current to decrease, resulting in a buck signal from the voltage reference and reduced output from the underexcited reactive ampere limit respectively. The system gain is such that when these changes are in equilibrium there is only a slight increase in underexcited reactive current beyond the limit start point.

The limit amplifier consists of a single magnetic amplifier with a voltage compensated bias winding and stabilizing inputs from exciter voltage and its own output. The limit amplifier output is connected to the regulator amplifier through a Zener diode which biases the output-input characteristic of the limit amplifier so that there is zero output for zero input as shown in Figure 12.

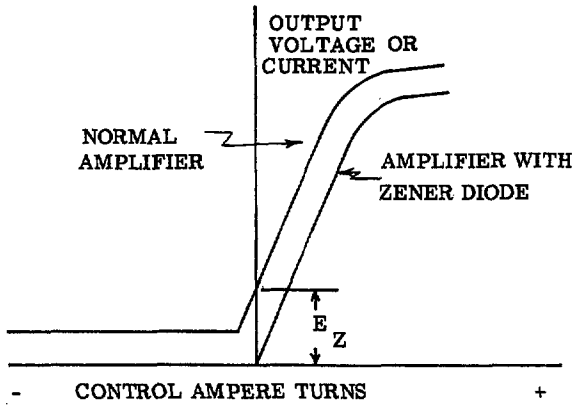


Figure 12. Characteristic of Biased Magnetic Amplifier Having Zener Diode in Output

Limit Amplifier Adjustments

The limit amplifier has coarse and fine bias adjustments used to set the output voltage at the rating of the Zener diode so that the output current start point corresponds closely with the input current start point. It has a voltage compensation adjustment which is set so that the start point is independent of the supply voltage throughout the expected range of supply voltage. There are also stabilizing and sensitivity adjustments set in accordance with system requirements. All adjustments are made in the factory, however it may be necessary to readjust the stabilizing adjustments at the time of installation due to variations in system parameters. The effect of the sensitivity adjustment is shown in Figure 13.

AUXILIARY EQUIPMENT

Active and Reactive Current Compensator

In certain special applications, an active and reactive current compensator is used to reproduce in miniature, at the regulator, the resistance and reactance between the AC machine and some predetermined point in the system. The function of the compensator is to lower the signal voltage to the regulator as the machine load increases, thereby causing the regulator to maintain normal voltage at some predetermined point on the system regardless of changes in the voltage drop over the system impedance, between the machine and this point.

In case line-drop compensation is required on machines that operate in parallel, provision may be made to prevent interference between the compensators. This is accomplished by special connections of the current transformer and compensators.

The reactance of the compensator is divided into a number of equal sections. Taps are brought out from these sections and connected to independently adjustable tap switches, one having coarse adjustments and the other fine. The reactance is variable in one-volt steps based on five amperes in the current transformer. It has a total of 24 steps, and will introduce a maximum of 24 volts total drop in the regulator signal voltage circuit with five amperes in the current-transformer secondary. This voltage drop is proportional to the current flowing in the compensator. The resistance portion of the compensator is a 5-ohm rheostat which produces 25 volts drop in the regulator signal with five amperes in the C. T. secondary.

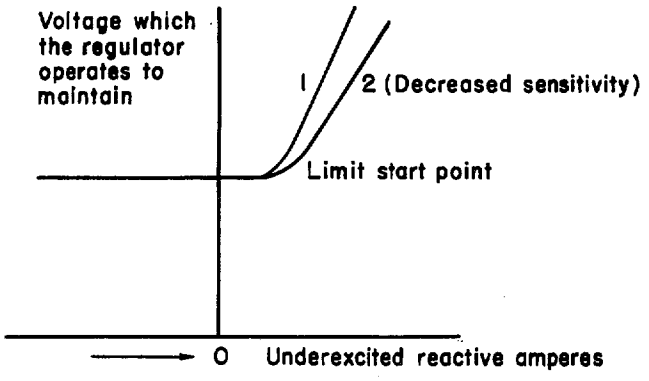


Figure 13. Reactive-Ampere Limit Sensitivity Adjustment Characteristic

Figure 14 shows the connections for this device when used with a single machine. Insulating windings are provided for both the reactor and resistor to permit grounding the current and potential-transformer secondaries separately. It is essential that the current transformer be in the same phase as the signal-voltage line in which the compensator is connected.

Amplidyne Generator

The amplidyne generator is normally individually driven by an induction motor, although other arrangements are sometimes used. Separate instructions are supplied for the amplidyne.

The amplidyne is usually provided with five control fields. The field and armature leads are marked with metal identification tags which should not be removed even after installation is complete. The field leads are connected to a small terminal board mounted on the machine frame, the terminals being marked with the field-lead identification numbers.

Two fields are excited by the regulator, and are so connected that the difference between their respective currents will determine the amplidyne voltage.

A single field is sometimes used as a negative-feedback winding to increase amplidyne linearity and speed of response. The output voltage of the amplidyne causes a current to flow through resistors D4R and the negative feedback field in a direction to decrease the amplidyne voltage (Figure 20).

The current rating and resistance of each field is indicated on the amplidyne nameplate.

The armature leads are marked A1 and C2 and the polarity of the leads will depend upon the magnitude and direction of current flow through the fields. Current flow through any field into the lowered numbered terminal and out the higher numbered terminal will cause armature terminal C2 to become positive. The effect of independent currents flowing in individual control windings is determined by the algebraic sum of the control ampere turns contributed by each winding.

Overhung on the amplidyne generator is a multi-phase, 200-cycle, permanent magnet alternator which supplies power for operating the regulator amplifier and obtaining an output from the regulator panel when there is no input from the machine potential transformers.

A flywheel is included in the M-G set to sustain speed on momentary interruptions of power and provide performance comparable to that obtained from high inertia

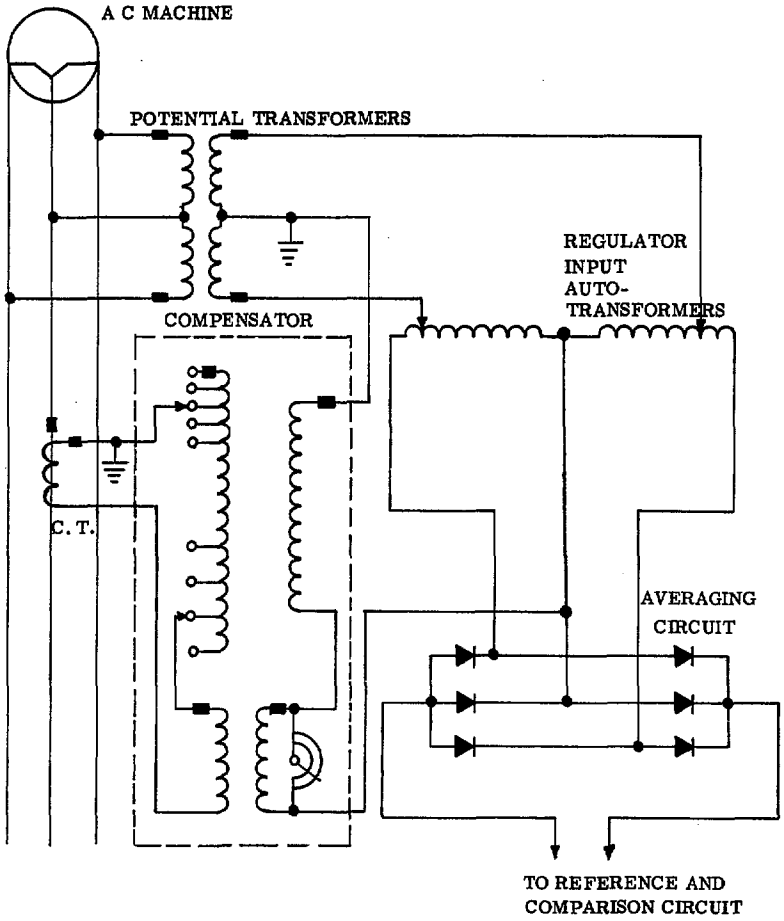


Figure 14. Active and Reactive Current Compensator Connections

exciter M-G sets.

CONTROL EQUIPMENT

The regulator equipment usually includes the components required for the control of the regulator and the amplidyne generator. Multicontact, dead-front, switchboard-type, cam-operated control switches are used for control of the regulator circuits. These switches are used for transfer and control functions, and may be supplied with fixed handles, or with removable handles where protective interlocking is required.

A motor starter is provided for the amplidyne driving motor. This starter may be equipped with a fused safety switch or a circuit breaker. The starter magnetic switch is a three-pole, DC operated device with overload relays. The operating coil may be rated for 125 or 250 volts DC.

A transfer contactor is provided for switching the amplidyne into and out of the exciter field circuit. This contactor may be a three-pole device with two normally open and one normally closed contact; or it may be two or three separate contactors. In the latter instances the contactor operating coils are series connected.

To protect against the possibility of establishing a serious amplidyne short circuit through the normally open and normally closed contacts of the transfer contactor, a current-limiting resistor is supplied for connection in this short-circuit path. This resistor is usually composed of several resistance units assembled with mounting brackets.

For Hydro applications, overspeed relay HSR and band pass filter H1X, H1C and H2C remove the regulator from service at about 140% overspeed.

The voltage adjusting rheostat is normally operated manually but may be motor operated, in which case it is located at the regulator rather than on the control board.

Instrumentation

Meters are normally provided for installation and maintenance purposes and meter the following quantities; average line voltage as seen by the regulator, regulator output current, amplidyne buck and boost currents, amplidyne output, U. R. A. L. output current and limit amplifier output current. An additional amplidyne meter is supplied for installation on the customer's control board.

P. T. and C. T. Burdens

The regulator and under-excited reactive ampere limit circuits impose a burden on the machine potential transformers which is approximately balanced at 1. 0 amperes in each phase at 120 volts in the case of steam turbines or synchronous condensers and 1. 0 amperes in each phase of hydro-generators.

The current transformer burden for all regulators is the same. The under-excited reactive ampere limit circuit imposes a burden of about 250 volt-amperes on the current transformer with 5 amperes flowing in the secondary. The reactive current compensator imposes a burden of 7. 5 volt-amperes on the current transformer at five amperes secondary current, and with all the reactance cut in. Approximately 1. 56 volt-amperes is the burden encountered in actual practice since it is usually necessary to turn in only a few steps of the equalizing reactor for successful parallel operation. The active and reactive current compensator imposes a maximum burden of 7. 5 volt-amperes with five amperes in the current transformer secondary and with the total reactance and resistance in the circuit.

INSTALLATION

LOCATION AND MOUNTING

The regulator components are usually supplied in an enclosing case designed for floor mounting. This enclosure should be installed in a well-ventilated, clean, dry location where the normal ambient temperature is not greater than 50C (122 F).

All of the equipment should be readily accessible for adjustment and testing. The amplidyne should be located to permit easy inspection and maintenance.

CONNECTIONS

Connections must be made in accordance with the diagrams supplied with the equipment for each particular installation. Care must be exercised to determine that the connections are correct to avoid damaging the equipment. Interconnecting wires conduct a maximum current of approximately five amperes in all regulator circuits. Maximum current in the amplidyne driving motor lines and amplidyne armature connections will be appreciably greater and will vary depending upon the ratings of these units.

Amplidyne driving motors may in some cases be connected for either of two voltages and the proper connections for the appropriate voltage must be made to avoid damaging the motor. Rotation of the amplidyne should be clockwise from the commutator end in the proper direction. Current transformer connections to the regulator should be made through a shorting device and exciter armature stabilizing connections through a disconnect device.

INITIAL OPERATION, TESTING AND ADJUSTMENT

CONTROL CIRCUITS

Control circuits should be tested under such conditions that the regulator will not affect the operation of the AC machine. First, turn the regulator control switch to OFF and energize the control-power supply. Under this condition the amplidyne-driving motor-starter magnetic switch, and the transfer contactor should remain deenergized. The regulator-control switch then may be turned to TEST. The motor-starter magnetic switch should be energized and the switch should close but the transfer contactor should remain in the deenergized position. The switch may then be turned to ON, the transfer contactor should be energized and should operate to place the amplidyne in the exciter field circuit. The switch may be returned to TEST to determine that the transfer contactor will be properly deenergized and the motor-starter magnetic switch will remain closed. Turning the switch to OFF should deenergize the magnetic switch.

For Hydro applications the overspeed relay circuit should be checked after placing the control switch at ON to energize the motor-starter magnetic switch and transfer contactor. Apply 28 volts DC to the coil of the overspeed relay. This should deenergize the motor starter and transfer contactor. The control switch should be left at ON and the coil deenergized. The devices should not be re-energized.

Some installations involve two or more regulator equipments consisting of normal equipments and spare equipments for a single or several AC machines. In such instances, transfer switches are frequently provided for selection of the proper regulator. If these switches are equipped with removable handles for protective interlocking, all handles but one of each type must be immediately removed and stored in a remote location to prevent incorrect and dangerous operation of the regulator equipment. The previously described tests must then be conducted on each equipment by turning the transfer switches to the proper positions to place the regulators successively under control of the respective control switches.

Regulator control circuits are interlocked with the exciter circuit breaker opening and closing control circuits, either with or without the use of relays. These circuits must be carefully tested by operating the circuit-breaker control switches and the circuit breakers themselves to ascertain that all details of

operation are correct.

In general, control circuits are arranged to prevent placing two regulators in service with one AC machine and exciter, and to prevent placing a single regulator in service with an exciter operating in parallel with a second exciter on the same AC machine field. The regulator diagrams must be carefully studied to determine what control functions are provided, and no further tests should be made until the control circuits have been completely tested.

VOLTAGE REGULATOR

Preliminary

The regulator should be tested with the AC machine operating at normal speed and normal voltage, and in the case of a generator, at no load and disconnected from the system.

The underexcited reactive ampere limit amplifier should be removed from service by disconnecting the wire from terminal B7, Figure 20.

The reactive current compensator knobs should be turned to the zero position and also the knobs of the active and reactive current compensator, if furnished.

Voltage Adjusting Unit

The regulator control switch should be kept at OFF or MANUAL until the AC machine has been started and brought up to normal speed and voltage. The exciter polarity should be checked to be certain that it is correct. After the control power supply has been energized, the switch may be turned to TEST to energize the regulator circuits. When this is done, the amplidyne armature voltage may suddenly increase in either direction. It should be possible to reduce the amplidyne voltage to zero by turning the voltage-adjusting unit. Turning the voltage-adjusting unit knob or voltage adjusting control switch clockwise should cause the amplidyne voltage to increase in the boost direction so that the amplidyne-voltmeter needle will move up scale from zero and amplidyne terminal C2 will become positive with respect to A1. Turning the voltage-adjusting unit knob or voltage-adjusting control switch counterclockwise should cause the amplidyne voltage to increase in the buck direction, so that the voltmeter will read down scale from zero and amplidyne terminal A1 will become positive with respect to C2. If the operation of the voltage-adjusting unit must be reversed, it may be changed as follows after the regulator control switch has been returned to OFF.

CAUTION

Do not allow the amplidyne to operate continuously at more than rated voltage.

The external connections to the two terminals of the voltage-regulator panel which connect to the ends of the voltage-adjusting unit may be interchanged. (The connection from the voltage-regulator panel to the slider on the voltage-adjusting unit should not be changed.) The control switch should then be turned to TEST and the operation of the voltage-adjusting unit rechecked to determine that it is satisfactory.

Voltage Level

It is possible that operation of the voltage-adjusting rheostat or switch will have no noticeable effect on the amplidyne voltage when the regulator is first tested.

Inability to control the amplidyne voltage may be due to incorrect signal voltage; if the amplidyne voltage is too high in the boost direction the signal voltage may be too low. The regulator is normally connected for a nominal signal voltage of approximately 115 volts but usually can be adjusted for nominal signal voltages from 85 to 140 volts by use of the taps on the autotransformers. The voltage-adjusting rheostat will usually provide adjustment of 7.5 percent above to 7.5 percent below the nominal signal voltage for steam turbines and + 10% to - 25% for hydro-generators. The signal-voltage supply should be connected to autotransformer taps which will cause the percent average line voltage meter to read closest to 100%. A higher signal voltage requires the use of higher numbered taps. Taps of the same number should be used on each transformer.

WARNING

Change taps on autotransformer only when the regulator control switch is set at "OFF."

Final connections should be made which will permit adjustment of AC machine terminal voltage over the desired range by use of the voltage-adjusting rheostat.

It is usually preferable to delay the final voltage-level adjustment and "Per Cent Average Line Voltage" meter adjustment until the underexcited reactive-ampere limit has been reconnected, since it may affect the voltage-adjusting rheostat setting for a given voltage. In addition, the reactive current compensator, and active and reactive current compensator may appreciably alter the signal voltage after they have been adjusted. Readjustment may also be required following the addition or removal of compensators. The voltage adjuster range may be extended to lower voltages by decreasing A1R and to higher voltages by decreasing A2R.

Additional Control-Switch Circuits

Where additional control, transfer, or exciter-polarity reversing switches are used, the previous tests should be repeated, with these switches adjusted for all operating conditions, to determine that the amplidyne voltage may be properly controlled with the voltage-adjusting unit when each of the regulator control switches is at TEST. This will indicate that all switch connections are correct.

If line voltage is near 100% and the amplidyne cannot

be controlled, it is possible that the amplidyne has a negative feedback circuit connected incorrectly. If so, disconnecting this circuit will make it possible to control the amplidyne in the test position, after which the connections to the negative feedback field should be reversed, reconnected, and retested.

Polarity Tests

Before the regulator is first placed in control of machine excitation, it is recommended that final polarity test be conducted. With the regulator control switch at OFF, disconnect the incoming leads to amplidyne control field F1-F2 and short them together. Do the same with the incoming leads to F3-F4. Connect a 1-1/2 volt dry cell to F1-F2 through a 200 ohm rheostat. With the control switch in the TEST position, set the rheostat to produce about 10 volts positive at amplidyne terminal C2. Place the regulator control switch in the ON position and observe the change in exciter voltage. Return the control switch to the OFF position.

If the exciter voltage increased when the switch was in the ON position, reconnect the control fields and proceed to the next step. If the exciter voltage decreased, determine the cause and correct it before proceeding.

It is also necessary to check the stabilizing connections. With the control switch at TEST and the AC machine operating at normal speed and voltage, set the voltage adjusting rheostat for zero amplidyne volts. Disconnect the exciter voltage from the regulator amplifier and short the stabilizing input terminals - not the exciter. The amplidyne voltage should go positive or in the boost direction. Remove the short and reconnect the exciter voltage input. If the amplidyne went negative or in the buck direction, locate and correct the trouble before proceeding.

The reactive current compensator polarity must be checked before putting the regulator in service with an AC machine which is connected to the system. Turn the compensator adjusting knobs to the zero position and operate the machine overexcited at normal voltage. With the regulator-control switch at TEST, adjust the amplidyne voltage to zero with the voltage-adjusting rheostat. Turn the compensator coarse-adjusting knob to the right to insert reactance. The amplidyne voltage should increase in the buck direction. If the voltage increases in the boost direction, the current transformer leads to the compensator must be reversed and the tests repeated. When the polarity of the compensator is correct, turn both knobs to zero and set the fine-adjusting knob at 5. It is then possible to proceed with the regulator tests. The amplidyne should go boost if the preceeding test is made with an active and reactive current compensator.

Initial Operation

It is recommended that the following tests be made with the AC machine disconnected from the load or system; if the AC machine must be connected to the system, it should be operated at a light load. For complete information on placing the regulator in service and removing it from service, refer to the section titled "Operation".

With the AC machine operating at normal speed and voltage and with the regulator-control switch at TEST, adjust the amplidyne voltage to zero. The control switch may then be turned to ON. This will place the amplidyne in the exciter-field circuit and the regulator in control of the AC machine excitation. The amplidyne voltage should not change more than about 20 volts and should immediately return to nearly zero. If the amplidyne voltage changes appreciably, immediately turn the control switch to OFF and determine the cause if incorrect operation.

A sudden and large change in amplidyne voltage may mean incorrect amplidyne or compensator polarity. Polarity check should be repeated before again placing the regulator in control of the AC machine excitation.

If the amplidyne exhibits a gradual change in voltage resulting in oscillations which grow steadily greater, the stabilizing connections or adjustment of the stabilizing circuits may be incorrect. The polarity of stabilizing circuits should be rechecked. If after all polarity connections have been rechecked, the amplidyne voltage oscillates when the control switch is placed at ON, remove the regulator from service. Now attempt to obtain stable operation by adjusting the resistor-rheostat combination in series with the capacitor. Again place the regulator in control of the AC machine excitation. If the stability has not improved, reduce the resistance to a value that is about 15 percent less than the original value and again place the regulator in control of the AC machine excitation. If the stability still has not improved, repeat the preceding adjustment using larger and smaller values of resistance. It should be possible to stabilize the regulator by gradual adjustments of this resistor-rheostat combination. However, if the regulator still is unstable after the full range of resistance has been tried, the exciter stabilizer capacitors should be added and disconnected one at a time. (Refer to the connection diagram supplied with the equipment.) After each change the resistance should be adjusted in the same manner as previously described.

WARNING

All adjustments must be made with the control switch at "OFF".

In general, stability difficulties are more likely to appear at reduced excitation than at normal AC machine excitation. Therefore, it is usually possible to obtain satisfactory stability under loaded conditions, if satisfactory stability is obtained at no load.

CAUTION

Satisfactory stability must be secured before further tests are conducted.

With the regulator-control switch at ON, turn the voltage-adjusting knob or switch in the raise direction; the amplidyne voltage should change slightly in the boost direction and the excitation armature voltage and AC machine terminal voltage should rise. Now turn the voltage-adjusting knob or switch in the lower direction; the amplidyne voltage should change in the buck direction and the exciter and AC machine terminal voltage should fall. If the AC machine is a generator which is not connected to a load or a system, the voltage-adjusting rheostat may be operated to obtain a total change in machine voltage of about 15 percent of rated voltage.

Operation of the voltage regulator may also be tested by changing the resistance of the exciter-field rheostat. As the rheostat resistance is decreased, the amplidyne voltage should change in the buck direction; as resistance is increased, the voltage should change in the boost direction. No appreciable change in exciter voltage should occur.

Refer to the section titled "Operation" for instructions on removing the regulator from control of machine excitation.

Where additional control, transfer, or exciter-polarity reversing switches are used, the previous tests should be repeated with these switches adjusted for all operating conditions to determine that the AC machine excitation may be properly controlled with the regulator. This will assure the correctness of all switch connections.

When proper regulator operation has been secured, optimum stability should be checked and the necessary adjustments made.

Optimum Regulator Stability

The following test should enable a quick determination of optimum stability. Place the regulator in control of AC machine excitation.

After suddenly turning the voltage-adjusting rheostat a few degrees in the lower direction, the amplidyne voltmeter will indicate only two or three oscillations and the AC machine voltmeter will overshoot only slightly before returning to a new steady value.

Regulator Sensitivity and Voltage Regulation

Determination of AC machine voltage regulation with the regulator in service is a difficult procedure under usual operating conditions and one which will produce only qualitative results. Since the regulator is adjusted at the factory to provide adequate sensitivity for close regulation, this measurement is usually unnecessary at the time of installation, and for this reason no special test procedure is given. After the equipment has been placed in service, it is possible to obtain data which will provide a measure of voltage regulation, but results must be carefully interpreted to gain a reliable estimate of performance.

If the machine is connected to a system, the regulation will depend to a great extent upon the characteristics of this system. Regulation will also be considerably affected by the use and adjustment of compensators. Furthermore, the sensitivity of the regulator itself will be a major factor affecting voltage regulation. A method is given in a later section for measuring regulator sensitivity.

Under-Excited Reactive Ampere Limit

Limit Polarity

After satisfactory operation of the regulator has been obtained, the reactive ampere limit should be tested. Reconnect the limit amplifier to the regulator amplifier. Set the REACTIVE AMPERE LIMIT POWER RECALIBRATION switch at zero and the REACTIVE AMPERE LIMIT START dial at its highest numbered position.

With the machine carrying power load and some safe value of under-exciter reactive current, move the regulator control switch to TEST. Adjust the voltage adjusting rheostat for zero amplidyne volts. Slowly turn the REACTIVE AMPERE LIMIT START dial towards zero. At some setting of the dial, the limit detector meter reading will go to zero and increase in the opposite direction. The limit signal current will increase from zero and the amplidyne output will go in the boost direction. If the limit detector current does not reverse before the dial has been turned to zero, return the dial to the highest numbered position. Decrease the AC machine excitation to further increase the under-excited current, being careful not to exceed the safe operating limit for the machine. It may be desirable to reconnect the limit to operate on over-excited reactive current as described in "Reactive Ampere Sensitive Circuit" in a previous section.

Again turn the REACTIVE AMPERE LIMIT START dial towards zero until the limit detector current is 10 MA. If the signal cannot be increased from zero by turning the LIMIT START dial to zero, the limit polarity may be reversed. Turn the regulator control switch to OFF and reverse the primary connections of transformer B3T (Figure 20). Repeat the test previously described to determine if reverse limit detector current can be obtained by turning the limit start dial towards zero. THIS TEST MUST GIVE PROPER RESULTS BEFORE FURTHER TESTS ARE CONDUCTED. If the limit signal current had an initial value that went to zero as the LIMIT START dial setting was reduced, the input to the limit amplifier must be reversed. If the amplidyne voltage increases from zero in the buck direction, the limit DC output is reversed. Reverse the connections between the limit amplifier and the regulator amplifier. Repeat the previously described test to secure proper results.

With the REACTIVE AMPERE LIMIT START dial so set that the amplidyne output is slightly in the boost direction, turn the POWER RECALIBRATION tap

switch from zero toward point 9. If the AC machine is delivering power, the amplidyne voltage should increase in the boost direction as the POWER RECALIBRATION switch is turned toward point 9.

Turn the POWER RECALIBRATION REACTOR switch to zero. Readjust the amplidyne voltage to 20 volts boost with the REACTIVE-AMPERE-LIMIT START dial. Before proceeding further with the tests on the underexcited reactive limit circuit, it is necessary to check the polarity of the reactive-ampere-limit stabilizing circuit. Carefully disconnect one of the exciter voltage stabilizing leads from the limit amplifier and short the stabilizing input terminals on the limit amplifier (refer to the connection diagrams supplied with each equipment). This should cause the amplidyne voltage to increase in the boost direction. Replacing the primary lead should cause the amplidyne voltage to change in the buck direction. Do not proceed with the underexcited reactive-ampere-limit tests until the polarity of the limit stabilizer is correct.

Initial Operation

Turn the REACTIVE-AMPERE-LIMIT START dial to its highest reading. Readjust the amplidyne voltage to zero with the voltage-adjusting rheostat, if necessary. Operate the AC machine at normal voltage and with underexcited reactive current. Turn the regulator control switch to ON as discussed under "Initial Operation". Slowly turn the REACTIVE-AMPERE-LIMIT START dial toward zero. The amplidyne voltage should increase slightly in the boost direction and the exciter voltage should increase causing the underexcited reactive current to decrease. The dial setting at which the underexcited reactive current starts to decrease is the limit-start point. If the operation is not as described, immediately remove the regulator from control of the AC machine excitation by turning the regulator-control switch to OFF. Repeat the limit polarity tests. Do not proceed further until satisfactory operation is obtained.

With the regulator-control switch at ON, and the REACTIVE-AMPERE-LIMIT START dial at the limit-start point, observe the amplidyne voltmeter, the exciter-voltmeter, and the AC machine ammeter for signs of oscillation. If oscillations of the reactive current or amplidyne voltage appear, remove the regulator from control of the AC machine excitation by turning the regulator control switch to OFF. Adjust the resistance in series with the exciter limit stabilizing capacitors in 15 percent steps, first in the direction to decrease resistance and then in the direction to increase resistance. After each of these adjustments repeat the procedure for putting the limit in service as previously described, being very careful to observe the amplidyne and exciter voltage oscillations and reactive current oscillations. However, if the limit still is unstable after the full range of resistance has been tried the exciter stabilizer capacitors on the limit amplifier should be added and disconnected one at a time. After each change the resistance should be adjusted in the same manner as previously described.

After stable operation of the limit has been obtained, check the limit operation as follows with the regulator in control of AC machine excitation, move the REACTIVE-AMPERE-LIMIT START dial to the limit-start point. Record the reactive current. Decrease the underexcited reactive-ampere load on the AC machine by operating the voltage-adjusting rheostat to raise the voltage. The underexcited reactive current should decrease. It should be possible to adjust the underexcited reactive current to any value lower than it was at the limit-start point. Now increase the underexcited reactive current, by turning the voltage-adjusting rheostat to lower the voltage. As the limit-start point is passed, the amplidyne voltage should increase in the boost direction and it should be impossible to raise the reactive current appreciably above the previously recorded value no matter how far the voltage-adjusting rheostat is turned in the direction to lower voltage.

As a final check of optimum limit stability, move the REACTIVE-AMPERE-LIMIT START dial until the amplidyne voltage is about 20 volts in the boost direction. Abruptly move the voltage-adjusting rheostat a few degrees in the lower direction and observe carefully the amplidyne voltmeter and the reactive-current ammeter for signs of oscillations. If oscillations appear, adjust for optimum stability as previously outlined. When the amplidyne voltmeter and the reactive-current ammeter show only a few oscillations after an abrupt change of the voltage-adjusting rheostat, the limit stability is satisfactory. Move the REACTIVE-AMPERE-LIMIT SENSITIVITY dial slowly toward its highest scale reading. Again check the limit stability.

This completes the preliminary adjustment of the limit. Final adjustment can be made at any time.

Final Adjustment of Limit

The final adjustment of the limit may be made by use of Figure 15 and 16, unless special calibration data are supplied with the equipment.

The limit-start adjustment is determined from Figure 15. This graph shows the value of the machine current-transformer secondary underexcited reactive current which will cause the limit to operate as a function of dial setting and normal AC machine potential transformer secondary voltage. Values of voltages differing from those shown on the graph may be easily interpolated. The POWER RECALIBRATING REACTOR adjustment is shown in Figure 16. This curve shows the amount by which the limit-start point will be reduced below the limit-start adjustment as a function of machine current-transformer secondary active current for various values of the tap-switch (B1SW) setting.

The two following examples are given to illustrate the method of setting the limit.

EXAMPLE 1. It is required that the limit should start to function when the underexcited reactive-current input to the limit reaches four amperes and that the limit action be independent of the power

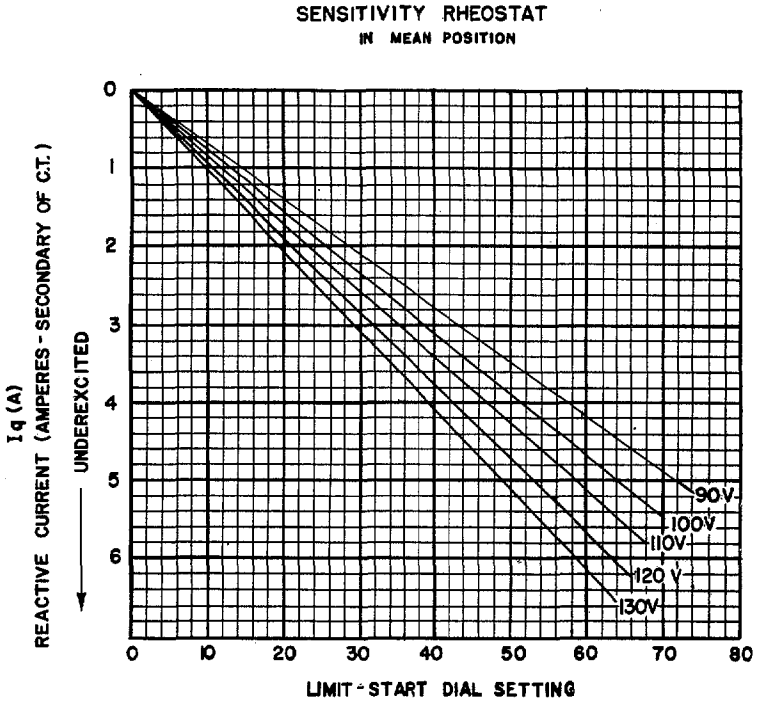


Figure 15. Calibration Curves for Reactive Ampere Limit Start Dial

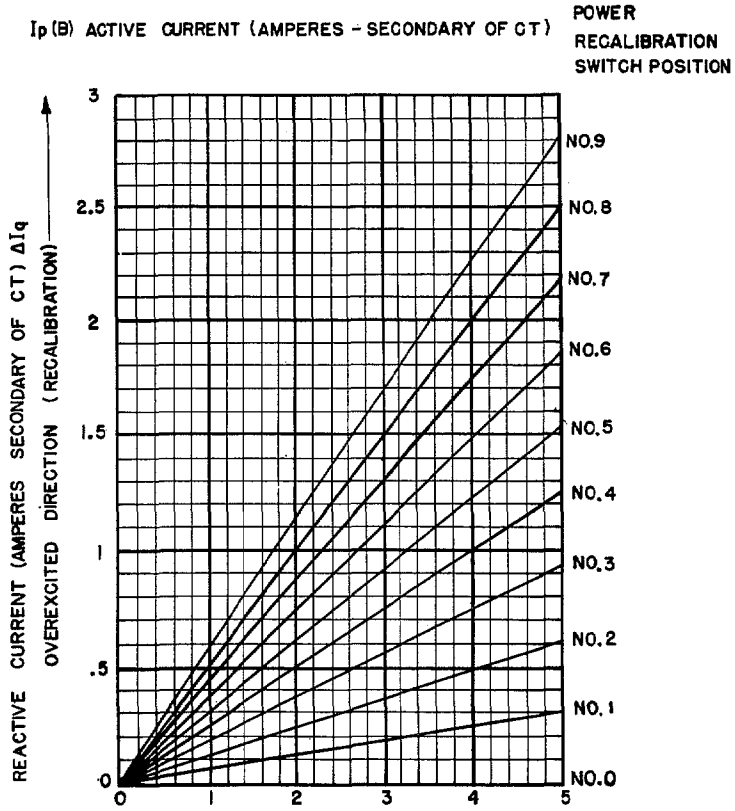


Figure 16. Calibration of Reactive-Ampere Power Recalibration Limit Circuit

component of current. The normal voltage on the secondary of the AC machine potential transformers is 110 volts.

Figure 15 indicates that for an underexcited reactive current of four amperes, the REACTIVE-AMPERE-START dial should be set at approximately 47. As can be seen from Figure 16, the tap switch on the power recalibrating reactor must be set on tap 0, since the limit action is to be independent of the power component of current.

EXAMPLE 2. It is desired to have the limit start to function when the underexcited reactive-current input to the limit reaches four amperes with zero active amperes, and when the underexcited reactive-current input reaches three amperes with four active amperes.

Figure 15 indicates that for an underexcited reactive current of our amperes, the REACTIVE-AMPERE-LIMIT START dial should be set at approximately 47. This satisfies the requirement at zero active amperes. Now, the number of reactive amperes necessary to start the limit at four active amperes must be reduced by one reactive ampere to obtain the desired power recalibration. Therefore, in Figure 16, a value of one reactive ampere recalibration of the limit and four active amperes to the limit, indicates that tap switch B1SW should be set at tap 4.

If desired, before making the final adjustment of the limit, the calibration curves (Figure 15 and 16) may be checked in the following manner.

Place POWER RECALIBRATING REACTOR tap switch B1SW on tap 8. Place the REACTIVE-AMPERE-LIMIT START dial at its highest scale position. Put the regulator in control of the AC machine excitation. With the AC machine carrying some convenient power load at about unity power factor, move the REACTIVE-AMPERE-LIMIT START dial slowly toward zero. At some position of the dial, the limit will start to operate. This position will be that which will just start to decrease the machine underexcited reactive current, or increase the overexcited reactive current.

Determine the active and reactive amperes delivered by the machine current transformer to the limit. Draw a vertical line from the active-ampere scale point in Figure 16, to the curve for tap 8. Read the corresponding reactive amperes recalibration of the limit, and the setting of the REACTIVE-AMPERE-LIMIT START dial.

In Figure 15, use the dial setting and the proper AC voltage curve to determine the value of underexcited reactive current for which the limit is set. From this value, subtract the reactive amperes recalibration. The result should be essentially equal to the reactive amperes delivered to the limit if the test has been carefully conducted.

Set the limit adjustments to the points which are desired for final operation. If it is desired to check the adjustments, the following procedure may be followed.

Place the regulator in control of the AC machine excitation. Operate the machine at the desired power load and at a reactive load which should not cause limit operation. Turn the voltage-adjusting rheostat in the direction to lower voltage until the limit prevents further reduction in machine overexcited reactive current, or increase in underexcited reactive current. Determine the value of reactive current and active current supplied by the machine current transformer to the limit.

Knowing the AC machine voltage and the setting of the REACTIVE-AMPERE-LIMIT START dial, determine the underexcited reactive current setting from Figure 15. Determine the underexcited reactive ampere recalibration from Figure 16. The result of subtracting the value of underexcited reactive amperes read in Figure 16 from the value of underexcited reactive amperes read in Figure 15 will be essentially equal to the value of underexcited reactive current to the limit at the limit-start point.

Reactive Current Compensator

The polarity of the reactive current compensator is checked as described under "Initial Operation". Final adjustment of the compensator can only be made after considerable experience with the machine operating under control of the regulator. It is desirable to keep the amount of reactance used, to the minimum required for proper division of reactive kva between machines, to avoid excessive voltage regulation. As an initial adjustment, it is frequently desirable to turn the fine-adjustment knob to position 5 with the coarse knob at zero. Adjustments may be made with the compensator current transformer energized. When making adjustments, the AC machine power factor should swing toward unity as the reactance of the compensator is increased.

Active and Reactive Current Compensator

Turn both adjusting knobs to zero, and operate the AC machine at lagging power factor. With the regulator control switch at TEST and the amplidyne voltage adjusted to zero, turn the coarse reactance-adjustment knob to the right to insert reactance; the amplidyne voltage should increase in the boost direction. Return the reactance-adjusting knob to zero, and turn the resistance-adjusting knob to the right to increase resistance; this should also increase the amplidyne voltage in the boost direction. If the amplidyne voltage increases in the buck direction during these tests, the compensator current leads must be interchanged and the tests repeated.

Final adjustment of the compensator must be made on the basis of experience. Preliminary adjustment may be made in accordance with the known values of resistance and reactance for that portion of the system over which compensation is desired. If the voltage at the point which is to be compensated decreases as the power factor becomes more lagging and increases as the power factor becomes less lagging, more reactance, and possibly less resistance may be required. Adjustments may be made with the com-

pensator and regulator in service.

Gain Measurements

The regulator-amplidyne gain and voltage adjuster calibration should be checked at the same time with the regulator in the TEST position. With the machine carrying normal load at normal voltage and the regulator off the underexcited reactive ampere limit, the PERCENT AVERAGE LINE VOLTAGE meter should be adjusted to read 100%. Actual adjustments should be made with the control switch in the OFF position. The voltage adjuster should be operated to zero the amplidyne output. Record dial setting of the voltage adjuster. Lower machine voltage with the exciter field rheostat to 98% and record amplidyne output volts. Raise the machine voltage to 102% with the exciter field rheostat, and record amplidyne volts. The voltage adjuster should be calibrated over its complete range using the exciter field rheostat to vary the machine voltage to obtain zero amplidyne output for each setting of the voltage adjuster. The regulator amplidyne gain and voltage adjuster calibration should then be calculated per Figure 17.

Underexcited Reactive Ampere Limit Gain

The underexcited reactive ampere limit gain should be measured using the voltage adjuster calibration obtained in the preceding test. With the regulator in service and the machine slightly overexcited off the reactive ampere limit, the voltage adjuster dial setting should be decreased in small steps, recording reactive power and dial setting for each step. The reactive current in the secondary of the current transformer should be calculated for each value of reactive power. The results should be plotted in accordance with Figure 18. The gain of the underexcited ampere limit is the reciprocal of the curve slope in the limit area and has the units percent volts per reactive ampere. It may be necessary to reduce the limit start dial setting and increase the power recalibration switch setting in order to drive the machine into the limit with the voltage adjuster.

	% LINE VOLTS	AMPLIDYNE OUTPUT	VOLT. ADJ. DIAL	COMMENT
0	100	0	48	E.F.R. & V.A. *
1	98		48	E.F.R.
2	102		48	E.F.R.
3		0	0	E.F.R.
4		0	10	E.F.R.
5		0	20	E.F.R.
6		0	30	E.F.R.
7		0	40	E.F.R.
8		0	60	E.F.R.
9		0	70	E.F.R.
10		0	80	E.F.R.
11		0	90	E.F.R.
12		0	100	E.F.R.

* E.F.R. - Use exciter field rheostat
V.A. - Use voltage adjuster

VOLTAGE ADJUSTER CALIBRATION - AVERAGE VALUE

$$\frac{[(12) - (3)]\ \%\ V}{[(12) - (3)]\ \text{Dial Division}} = \underline{\qquad} \%\ \Delta\ V/\text{Dial Division}$$

Typical: $\frac{(110 - 90)\ \%\ V}{100 - 0\ \text{Dial Division}} = .\ 2\%\Delta V/\text{Dial Division}$

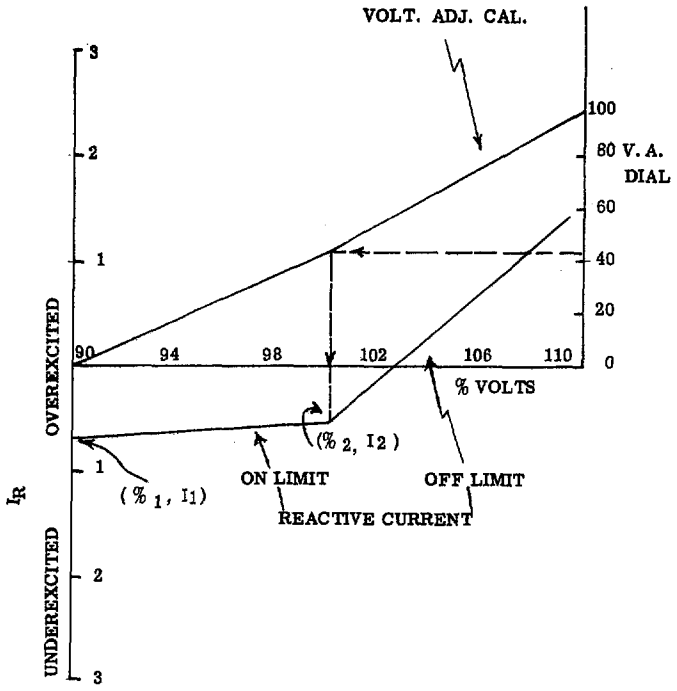
REGULATOR -AMPLIDYNE GAIN

$$\frac{[(2) - (1)]\ \text{V Amplidyne}}{[(2) - (1)]\ \%\ V} = \underline{\qquad}\ \text{V Amplidyne / \% V}$$

TYPICAL

$$\frac{(-160 - 140)\ V_A}{(102 - 98)\ \%\ V} = \frac{-300}{4} = -75\ \text{V Amplidyne / \% V}$$

Figure 17. Voltage Adjuster Calibration and Regulator Gain Measurements



VOLT. ADJ. DIAL	% V (A)	MVAR	IR (B)
70			
60			
50			
40			
30			
20			
10			
0			

(A) % V - Obtain from Fig. 21.

(B) I_R = Reactive amperes in C. T. Secondary

$$MVAR = (I_R)\ (C.T.\ RATIO)\ (E_T)\ (\sqrt{3})\ (10^{-6})$$

E_T = MACHINE TERMINAL VOLTAGE AT LIMIT START POINT

$$\frac{I_R}{MVAR} = \frac{10^6}{(C.T.\ RATIO)\ (E_T)\ (\sqrt{3})}$$

TYPICAL

$$\frac{I_R}{MVAR} = \frac{10^6}{(1200)\ (13,800)\ (\sqrt{3})} = 0.035\ \frac{AMPS}{MVAR}$$

LIMIT GAIN

$$\frac{\%}{I_R} = \frac{[\%_2 - \%_1]\ \%\ V}{[I_2 - I_1]\ I_R} = \frac{\quad}{\quad}\ \frac{\%\ V}{I_R}$$

TYPICAL

$$\frac{(100 - 90)\ \%\ V}{[-0.5 - (-0.66)]\ I_R} = \frac{10\ \%\ V}{0.16\ I_R} = 60\ \%\ V/I_R$$

Figure 18. U. R. A. L. Gain Measurement

OPERATION

NORMAL OPERATION

General

The complete voltage-regulator equipment should be placed in normal service with the AC machine only after the control circuits, regulator, underexcited reactive-ampere limit, reactive current compensator, and active and reactive current compensator have been properly tested in general conformance with the previously described instructions. Final adjustment of these units may, of course, be delayed until operating experience has been obtained, but circuits which have not been thoroughly tested must not be employed with the regulator in service if the possibilities of damage to the equipment and disturbance of the system are to be avoided.

Operation With Attended Equipment

In attended stations, the AC machine is generally brought up to normal speed and voltage under manual control before the regulator is placed in service. Generators are frequently connected to the load or system before the regulator is used, but this is not essential since the regulator may be placed in service with the machine under any load condition and with the machine connected to or disconnected from the system.

The following procedure should be used to place the regulator in control of machine excitation:

1. With the regulator-control switch at OFF, and with the AC machine operating at normal speed, adjust the machine terminal voltage to approximately normal with the exciter-field rheostat. If the machine is operating under loaded conditions or connected to the system, this step should not be necessary.

2. Turn the regulator-control switch to TEST.

3. Adjust the amplidyne voltage to zero with the voltage-adjusting rheostat or switch. If this cannot be done, do not proceed further until the trouble has been determined and eliminated.

4. Turn the regulator-control switch to ON. This will place the regulator in control of the AC machine excitation.

5. The terminal voltage or excitation level of the AC machine may now be adjusted with the voltage-adjusting rheostat or switch as required for the normal operating schedule.

The regulator will successfully control the AC machine excitation over the complete load range of the machine with the exciter-field rheostat at a position which will give rated machine voltage at no load without the regulator in service, or at a position which will give rated machine voltage at rated load without the regulator in service, or at any intermediate position. However, additional excitation system reliability may be-secured by proper adjustment of the exciter-field rheostat when the regulator is in control of machine excitation.

For this purpose, it is recommended that immediately after the regulator has been placed in service, the field rheostat be adjusted to and maintained at a predetermined position. This position should be such that the machine excitation will be sufficient under all normal AC machine loads to give stable operation and avoid serious operating disturbances if the excitation system should be suddenly returned to manual control through an emergency, resulting in loss of control by the regulator. It is suggested that this rheostat position be selected so that with the excitation system under manual control, it will produce from 3/4 rated to rated AC machine field current, with preference given to the latter value. Where the AC machine is closely paralleled with other machines equipped with automatic voltage regulators, without underexcited reactive ampere or minimum excitation limits, a position corresponding to the lower value may be found more desirable. Rheostats are normally supplied with a position switch which may be used in conjunction with remote indicating lights to facilitate obtaining the predetermined setting as a normal operating procedure. Operation with the rheostat at such a predetermined position should result in buck amplidyne voltage at most AC machine loads, and as a result, under most load conditions, sudden return to manual control will actually cause an increase in excitation.

Regardless of the operating procedure employed when the regulator is in control of the AC machine excitation, it is essential that under any sustained load condition a field-rheostat position be maintained which will give sufficient excitation in the event of sudden return to manual control, to permit stable operation in the case of a generator, and to avoid other serious disturbances for all types of AC machines.

The regulator may be removed from service and the AC machine excitation returned to manual control under any load condition. The following procedure should be used for this purpose:

1. Adjust the amplidyne voltage to zero by use of the exciter-field rheostat.

2. Turn the regulator control switch to OFF.

With the regulator in control, the exciter-field rheostat may be adjusted as follows to bring the amplidyne voltage to zero.

1. If the amplidyne voltage is in the boost direction, slowly increase the rheostat resistance until the amplidyne voltage is zero. No appreciable change in exciter voltage or AC machine voltage should occur.

2. If the amplifyne voltage is in the buck direction, slowly increase the rheostat resistance. No appreciable exciter or AC machine voltage change should appear.

Operation With Unattended Equipment

In unattended installations, special attention must be given to control circuitry to avoid exceeding permissible volts per cycle ratings of other equipment at sub-synchronous speeds. If regulator is in use in automatic mode, the generator terminal volts will be held constant as speed decreases. This will give high volts per cycle.

EXCITER POLARITY REVERSAL

Incorrect Polarity

If, when the AC machine exciter is started at any time, it is found that the exciter polarity is reversed, the polarizing circuit, if furnished, may be used to correct this condition. The exciter should be operated at reduced voltage and whenever possible, with the AC machine disconnected from the load or system. Turn the regulator control switch through INTERMEDIATE to POLARIZE. The amplidyne voltage should increase in the boost direction but the exciter

voltage should reduce to zero and then immediately increase with the proper polarity.

After the exciter polarity is correct, the control switch must be returned to OFF and the normal starting cycle completed.

MAINTENANCE

STATIC EQUIPMENT

The equipment should be kept relatively clean and dry. If vibration is present, all screw-type connections should be checked regularly to determine that they are properly tightened. Normally, the static components should require no further attention.

OTHER EQUIPMENT

Motor starter, amplidyne contactor, control relays, and other magnetically-operated contact making devices should be regularly inspected and maintained in accordance with applicable instructions for these devices. Normally the amplidyne will be the only unit requiring frequent inspection, and this machine should be maintained as described in the amplidyne instructions.

Variable transformer contact brushes should be inspected annually and the brushes should be reset by working them back and forth across the total winding surface many times. If arcing is present or if the brush becomes worn, a complete, new brush assembly should be installed. Since it is made of special material it should be obtained from the transformer manufacturer. In addition, where discoloration is present, clean the contact surfaces with crocus cloth. The same procedure should be followed with the voltage adjusting rheostat, and the sensitivity rheostats.

TROUBLE SHOOTING

GENERAL

In the event of abnormal operation of any part of the regulator equipment the complete equipment should be immediately removed from service and carefully tested to determine the cause of the difficulty. When incorrect operation is noticed, it is often possible to reduce greatly the overall servicing time by studying all of the symptoms. The trouble can usually be isolated by the process of elimination, if the circuit functions are clearly understood.

If any connections are changed, the affected circuits should be fully tested and adjusted before being returned to service.

TROUBLE SHOOTING CHART

TROUBLE	POSSIBLE CAUSE
Regulator operation apparently normal, except that regulator will not reduce excitation to the desired value.	Underexcited reactive ampere limit adjustment incorrect.
Sudden change in excitation and reduction of amplidyne voltage to zero (no control of amplidyne voltage possible with voltage-adjusting rheostat).	Opening of amplidyne driving motor starter switch due to loss of control power, operation of overload relays, operation of amplidyne overspeed switch, or failure of control switch.
Sudden increase in excitation and amplidyne boost voltage (no control of amplidyne voltage possible with voltage-adjusting rheostat).	Open connection in signal circuit due to loose connection, failure by open circuit of voltage-adjusting rheostat or other circuit component.
Sudden decrease in excitation and increase in amplidyne buck voltage (no control of amplidyne voltage possible with voltage-adjusting rheostat).	Open connection in current reference circuit or failure of any circuit component.
Sudden change in excitation (with control of amplidyne voltage possible by use of voltage-adjusting rheostat, but no control of excitation possible with control switch at ON).	Amplidyne contactor open due to coil failure, open control circuit, loss of control power, or control switch contact failure.
Failure of underexcited reactive-ampere limit to function at the proper values.	High resistance at brush contact surface in limit start variable transformer.

TROUBLE SHOOTING CHART (Continued)

TROUBLE	POSSIBLE CAUSE
Variation of machine power factor (with system or load power factor essentially constant).	Incorrect adjustment or failure of equalizing reactor or line-drop compensator.
Erratic fluctuations of amplidyne voltage and exciter voltage during operation of voltage-adjusting rheostat or limit.	Poor contact between brush and wire on voltage-adjusting rheostat.
Regulator stable with some amplidyne voltages, but unstable with other amplidyne voltages.	Improper operation of amplidyne.

RENEWAL PARTS

When ordering renewal parts, the following information should be given:

1. Catalog number stamped on the part, with a complete description, including use and location.

2. Complete nameplate data appearing on the assembly of which the part is a component.

3. If possible, data on original order on which equipment was first supplied, including all numerical references.

Renewal parts for the variable-transformer and motor drive for the voltage-adjusting potentiometer should be ordered directly from the manufacturer.

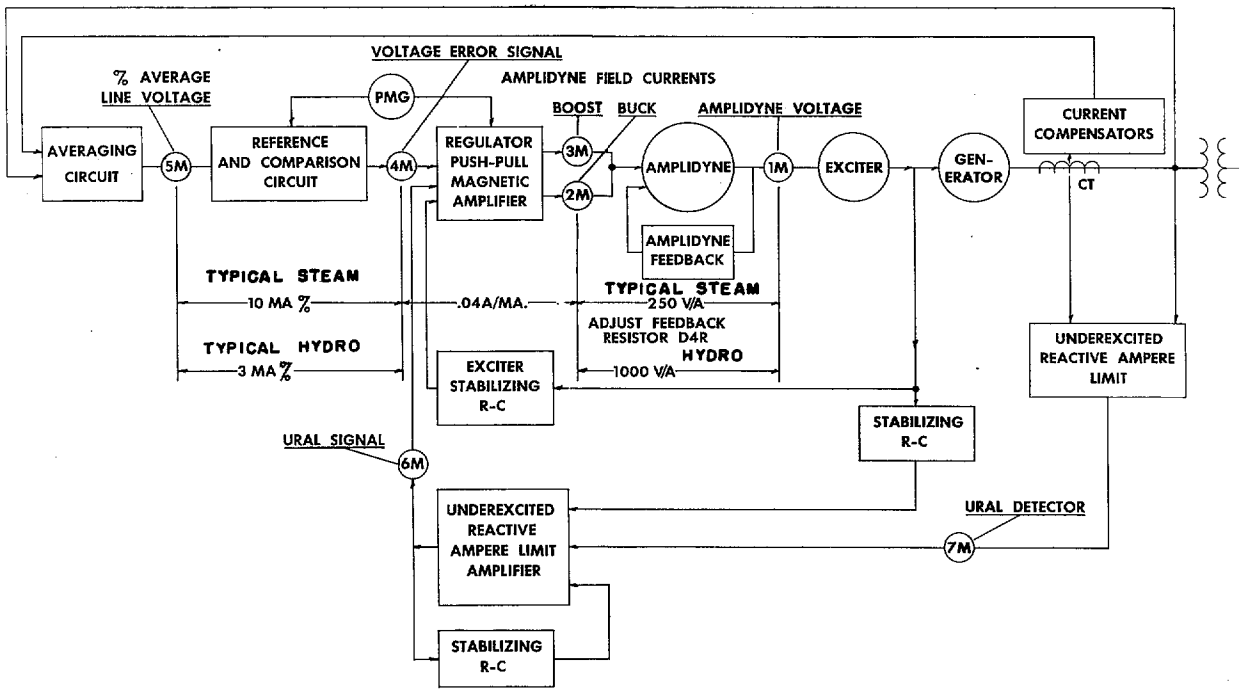


Figure 19. Block Diagram of Voltage Regulator

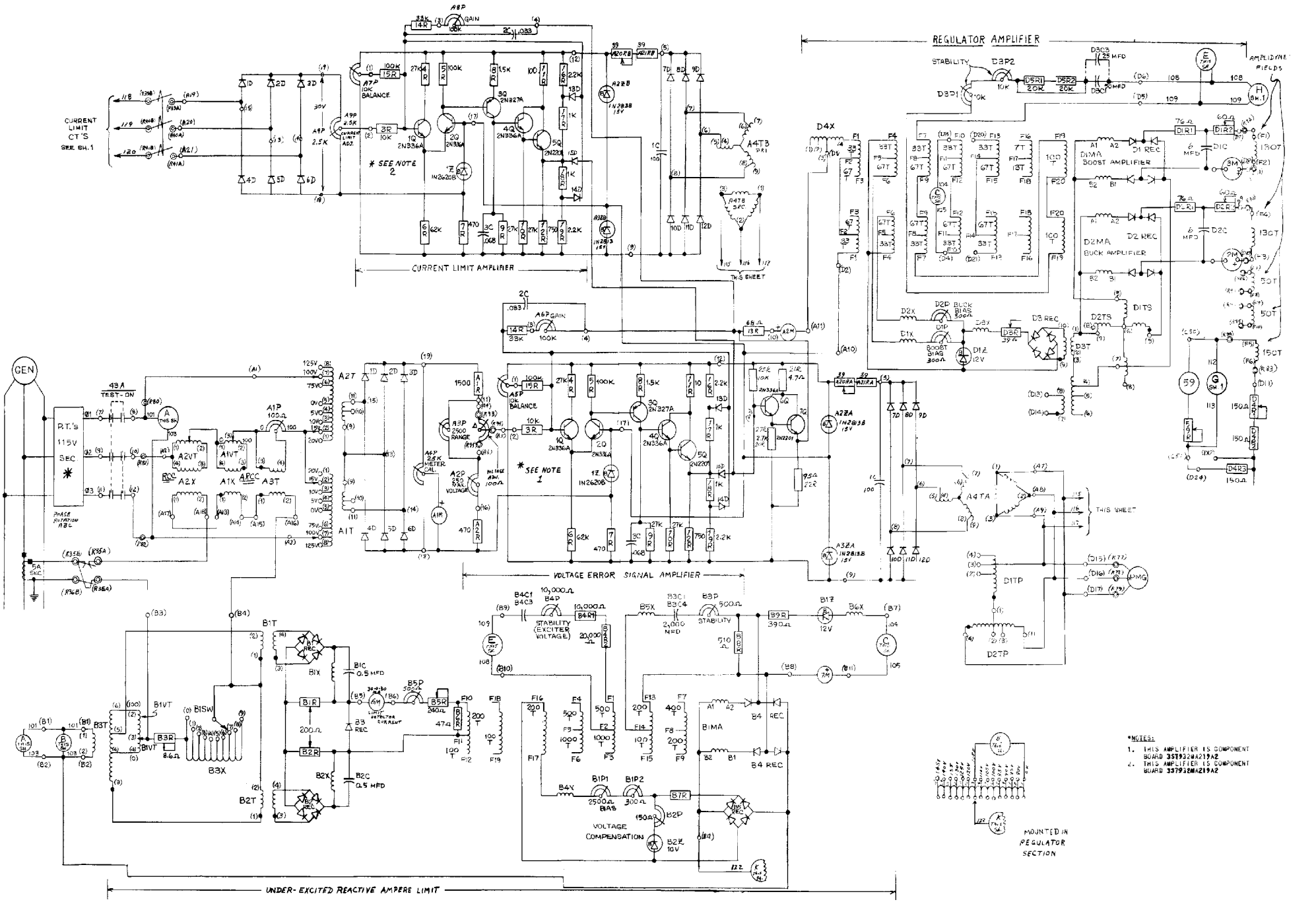


Figure 20A. Typical Elementary Diagram of Alterrex Excitation System

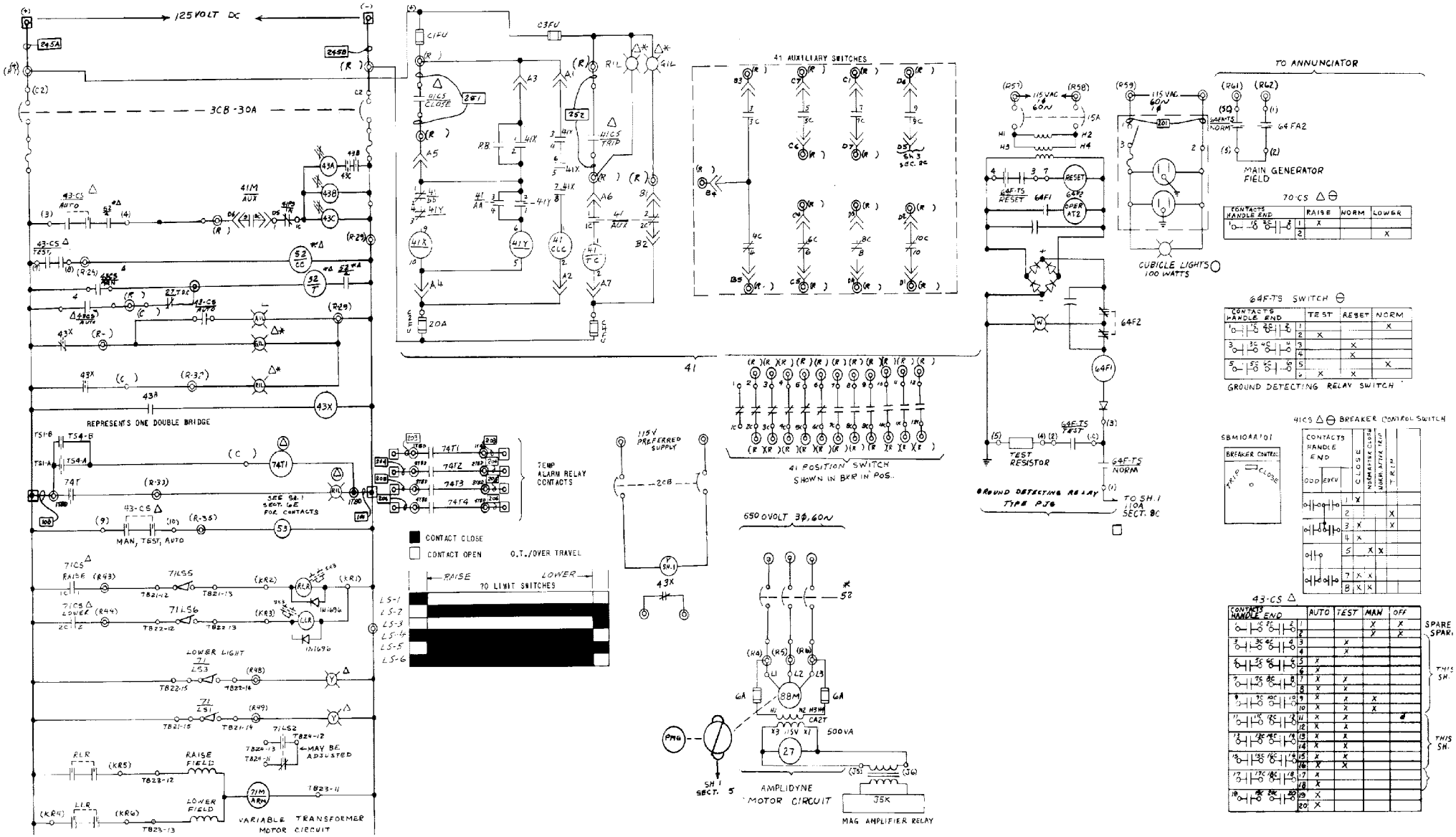


Figure 20B. Typical Control Circuits of Alterrex Excitation System